ServiceNow Integration Guide

This guide covers three ways to connect the ITIL Reflexion Agent to your ServiceNow instance, from simplest to most advanced.

Option 1: Export & Upload (No Integration Required)

The fastest path — export data from ServiceNow, upload it through the dashboard.

Exporting Incidents

1. In ServiceNow, navigate to **Incident > All**
2. Apply filters for the change you're documenting (e.g., category, CI, date range)
3. Click the **hamburger menu** (top-left of list) → **Export** → **JSON** or **CSV**
4. In the Reflexion dashboard, toggle to **Upload Your Data** and upload the file

Minimum fields needed: `number` (or `id`), `short_description` (or `title`), `priority` (or `severity`), `category`, `description`

The agent maps common ServiceNow field names automatically: | ServiceNow Field | Maps To | |—|—| | `number` | `id` | | `short_description` | `title` | | `priority` | `severity` | | `cmdb_ci` | `affected_ci` | | `close_notes` | `resolution` | | `sys_created_on` | `created` |

Exporting CMDB Items

1. Navigate to **Configuration > All CIs** (or a specific CI class)
2. Filter to the CIs relevant to your change
3. Export as JSON or CSV

Minimum fields: `name` (or `ci_id`), `sys_class_name` (or `type`), `short_description` (or `description`)

When to Use This

- Evaluating the agent for the first time
- One-off RFC generation
- No ServiceNow admin access available
- Security team hasn't approved API access yet

Option 2: ServiceNow REST API Token (Recommended for Production)

Direct API connection using a scoped service account. This is how most SaaS tools integrate with ServiceNow.

Step 1: Create a Service Account

In ServiceNow:

1. Navigate to **User Administration > Users**
2. Create a new user (e.g., `svc_reflexion_agent`)

3. Assign the role: `itil` (read access to incident, change, CMDB tables)
4. **Do NOT assign** `admin` — principle of least privilege

Step 2: Generate API Credentials

Option A: Basic Auth (simplest) - Use the service account username and password - Set in environment:
`SERVICENOW_INSTANCE=https://your-instance.service-now.com`
`SERVICENOW_USERNAME=svc_reflexion_agent` `SERVICENOW_PASSWORD=your-secure-password`

Option B: OAuth 2.0 (recommended for production) 1. In ServiceNow, navigate to **System OAuth > Application Registry** 2. Create a new OAuth application 3. Note the Client ID and Client Secret 4. The agent will use client credentials grant to obtain tokens

Step 3: Test the Connection

```
# Test that your credentials can read incidents
curl -u "svc_reflexion_agent:password" \
  "https://your-instance.service-now.com/api/now/table/incident?sysparm_limit=1"

# Test CMDB access
curl -u "svc_reflexion_agent:password" \
  "https://your-instance.service-now.com/api/now/table/cmdb_ci?sysparm_limit=1"
```

A `200` response with JSON data confirms your credentials and permissions are correct.

Step 4: Configure the Agent

Set the environment variable and restart:

```
export SERVICENOW_INSTANCE=https://your-instance.service-now.com
export SERVICENOW_USERNAME=svc_reflexion_agent
export SERVICENOW_PASSWORD=your-secure-password
python main.py
```

ServiceNow Tables Used

The agent queries these tables (read-only):

Table	API Endpoint	Purpose
`incident`	`/api/now/table/incident`	Incident history for risk analysis
`change_request`	`/api/now/table/change_request`	Past changes for context
`cmdb_ci`	`/api/now/table/cmdb_ci`	Configuration items and dependencies
`cmdb_rel_ci`	`/api/now/table/cmdb_rel_ci`	CI relationships (optional)

The agent **never writes** to ServiceNow. All access is read-only.

When to Use This

- Ongoing RFC generation against live data
- Production deployment
- When you want the agent to always have current incident/CMDB data

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Option 3: MCP Server Bridge (Advanced)

Connect via Model Context Protocol using a community MCP server running in your network.

Architecture

```
Your Network                          Cloud / Local
┌──────────────────────┐                ┌──────────────────────┐
│  ServiceNow Instance │                │  Reflexion Agent     │
│         ↓            │                │       ↓              │
│  MCP Server          │  ←── HTTPS ──→ │  MCP Client          │
│  (snow-mcp or        │                │  (tools.py)          │
│   servicenow-mcp)    │                │                      │
└──────────────────────┘                └──────────────────────┘
```

The MCP server runs inside your network with access to ServiceNow. The Reflexion Agent connects to it over HTTPS.

Setup with snow-mcp (60+ tools)

```
# On a server with ServiceNow network access
git clone https://github.com/ShunyaAI/snow-mcp.git
cd snow-mcp
npm install

export SERVICENOW_INSTANCE=https://your-instance.service-now.com
export SERVICENOW_USERNAME=svc_reflexion_agent
export SERVICENOW_PASSWORD=your-secure-password

npm start  # Default port 8200
```

Setup with servicenow-mcp

```
git clone https://github.com/michaelbuckner/servicenow-mcp.git
cd servicenow-mcp
pip install -r requirements.txt

export SERVICENOW_INSTANCE=https://your-instance.service-now.com
export SERVICENOW_USERNAME=svc_reflexion_agent
export SERVICENOW_PASSWORD=your-secure-password

python server.py  # Default port 8200
```

Configure the Reflexion Agent

```
export SERVICENOW_MCP_URL=http://your-mcp-server:8200
python main.py
```

Exposing the MCP Server Securely

If the Reflexion Agent runs outside your network (e.g., Cloud Run), you need a secure tunnel:

Cloudflare Tunnel (recommended):

```
cloudflared tunnel --url http://localhost:8200
```

ngrok (for testing):

```
ngrok http 8200
```

Then set `SERVICENOW_MCP_URL` to the tunnel URL.

How Data Flows

When `SERVICENOW_MCP_URL` is configured:

1. The `retrieve_data` node calls `search_incidents` via the MCP server
2. The MCP server queries ServiceNow's REST API
3. Results are returned as JSON to the agent
4. The agent processes real incident/CMDB data through the Reflexion loop

When NOT configured: - All data comes from bundled JSON fixtures in `data/` - Three demo scenarios work out of the box

Testing the MCP Connection

```
# Verify the MCP server is running
curl http://localhost:8200/tools

# Test incident search
curl -X POST http://localhost:8200/tools/search_incidents \
  -H "Content-Type: application/json" \
  -d '{"query": "database performance", "limit": 5}'

# Test CMDB lookup
curl -X POST http://localhost:8200/tools/get_cmdb_info \
  -H "Content-Type: application/json" \
  -d '{"ci_id": "DB-PROD-SQL01"}'
```

When to Use This

- You want real-time ServiceNow data but can't expose ServiceNow directly
- You need the full MCP tool catalog (60+ tools with snow-mcp)
- You're running the Reflexion Agent as part of a larger MCP ecosystem
- You want to connect Claude Desktop or other MCP clients to the same tools

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Option 4: Native ServiceNow MCP (Zurich Release+)

ServiceNow's Zurich release includes native MCP support built into the platform.

If your instance is on Zurich or later:

1. Enable MCP in your ServiceNow instance (System Properties)
2. Configure which tables/operations to expose
3. Set the MCP endpoint URL as `SERVICENOW_MCP_URL`

No third-party MCP server needed. Consult ServiceNow's documentation for your specific version.

Security Considerations

Principle of Least Privilege

- Create a dedicated service account — never use `admin`
- Grant only `itil` role (read access to incident, change, CMDB)
- Review and audit API access periodically

Network Security

- Use HTTPS for all connections
- If using MCP bridge, restrict access by IP or use authenticated tunnels
- ServiceNow supports IP access control lists (ACLs) for API access

Data Handling

- The agent processes data in memory only — nothing is persisted to disk
- No incident or CMDB data is sent to third parties (only to the configured LLM provider)
- When using cloud-hosted LLMs, be aware that incident descriptions are sent as prompt context

Compliance

- The agent is read-only — it never modifies ServiceNow data
- All generated RFCs are returned to the user, not written back to ServiceNow
- Audit trail: the streaming log shows exactly what data was retrieved and how it was used

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Troubleshooting

"Backend not available" in the dashboard

- Verify the Reflexion Agent backend is running: `curl http://localhost:8080/api/health`
- Check that `SERVICENOW_MCP_URL` is set correctly

MCP server returns empty results

- Verify ServiceNow credentials: test with a direct REST API call (see Step 3 in Option 2)
- Check that the service account has `itil` role
- Verify the MCP server logs for authentication errors

"Falling back to fixtures" message

- The agent couldn't reach the MCP server — check the URL and network connectivity
- This is a graceful fallback, not an error — the agent still runs using demo data

Slow queries

- ServiceNow REST API can be slow on large tables — add filters (date range, category) to narrow results
- The MCP server may need pagination configuration for large result sets

SSL/TLS errors

- Ensure your ServiceNow instance certificate is valid
- If using a corporate CA, the MCP server may need the CA bundle configured

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Data Field Mapping Reference

Incident Fields

ServiceNow Field	Agent Field	Required
`number`	`id`	Yes
`short_description`	`title`	Yes
`priority`	`severity`	Yes
`category`	`category`	Recommended
`description`	`description`	Recommended
`cmdb_ci.name`	`affected_ci`	Recommended
`close_notes`	`resolution`	Optional
`sys_created_on`	`created`	Optional
`business_duration`	`mttr_hours`	Optional

CMDB Fields

ServiceNow Field	Agent Field	Required
`name`	`ci_id`	Yes
`sys_class_name`	`type`	Recommended
`short_description`	`description`	Recommended
`business_criticality`	`criticality`	Optional

Fields not present in the upload are handled gracefully — the agent works with whatever data is available.